

NovaCred

Credit Application Governance Audit

Group 14

2606 - Data Ecosystems and Governance in Organizations

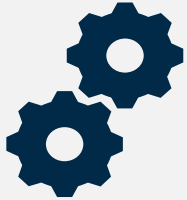
Nova School of Business and Economics | MSc Business Analytics

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Meet the Team

Group 14 Data Governance Task Force

Data Engineer



Xavier Albino

- Data loading
- Cleaning
- pipeline code

Data Scientist



Othmane
Chadi

- Bias analysis
- Fairness metrics
- Statistical testing

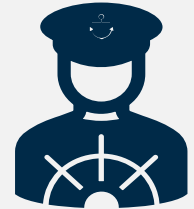
Governance Officer



Amgirus
Murali

- GDPR mapping
- Policy
- Recommendations
- Compliance analysis

Product Lead



Philipp
Connert

- Presentation
- Coordination
- README documentation

Our audit is organized into four pillars contributing to a systematic governance examination of NovaCred's credit application system

Overview of Sections

1 Data Quality Assessment

2 Bias & Fairness Analysis

3 Privacy & Governance

4 Recommendations



1

502 Credit Applications Across 21 Fields, Including Sensitive PII

Data Quality Assessment



Category	Key Fields	Sensitivity
Applicant Identity	full_name, email, SSN, DOB, gender, zip_code, IP address	HIGH / CRITICAL
Financial Profile	annual_income, credit_history_months, debt_to_income, savings_balance	MEDIUM
Spending Behaviour	spending_behavior (category + amount list)	HIGH (behavioural PII)
Loan Decision	loan_approved, approved_amount, interest_rate, rejection_reason	MEDIUM
System Metadata	application_id (_id), processing_timestamp, notes	LOW

1 Assessment across completeness, uniqueness, consistency, validity, accuracy, and timeliness. From 502 raw to 498 cleaned records.

Data Quality Assessment



- **14 hidden blank strings** undetected by standard null checks
- **87.6% of processing timestamps absent** - no audit trail
- SSN, DOB, email each missing in ~1% of records

Completeness



- **4 duplicate application IDs** found (0.8% of records)
- **3 SSNs shared** across different applicants - identity collision

Uniqueness



- **Gender** stored as M/F AND Male/Female - encoding **mismatch**
- **Date of birth** in 3 formats (ISO, DD/MM/YYYY, YYYY/MM/DD)
- **Income** in two field names (annual_income / annual_salary)

Consistency



- 2 records: **negative credit history** months (impossible value)
- 1 record: debt-to-income > 1.5 and 1 record: **negative savings**
- 4 **email** addresses fail **format validation** (e.g. missing @)

Validity



- Age range: 21–65 years - no implausible values detected
- Decision field integrity: 0 violations across all checks
- No approved loans with missing income (process gate intact)

Accuracy



- Only 12.4% of records carry a **processing timestamp**
- 2 timestamps set in the **future** (data entry error)
- Dataset recency cannot be assessed - governance blind spot

Timeliness

1 Quality Failures Create Direct Credit Decision and Compliance Risk

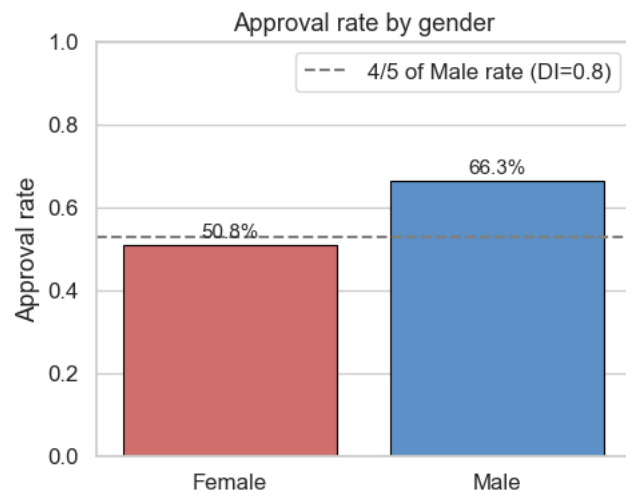
Data Quality Assessment

DATA DEFECT	BUSINESS RISK	REGULATORY CONSEQUENCE
Missing timestamps (87.6%)	No audit trail for decisions	GDPR Art. 5(2) accountability violation
3 duplicate SSNs	Wrong applicant decision possible	Art. 22 - automated decision on incorrect data
Mixed date/gender formats	ML model training inconsistency	EU AI Act Art. 10 - data governance failure
Invalid email/numeric values	Pipeline corruption risk	GDPR Art. 5(1)(d) or EU AI Act

Gender disparate impact (DI < 0.8) and age patterns indicate fairness risk in approvals

Bias & Fairness Analysis

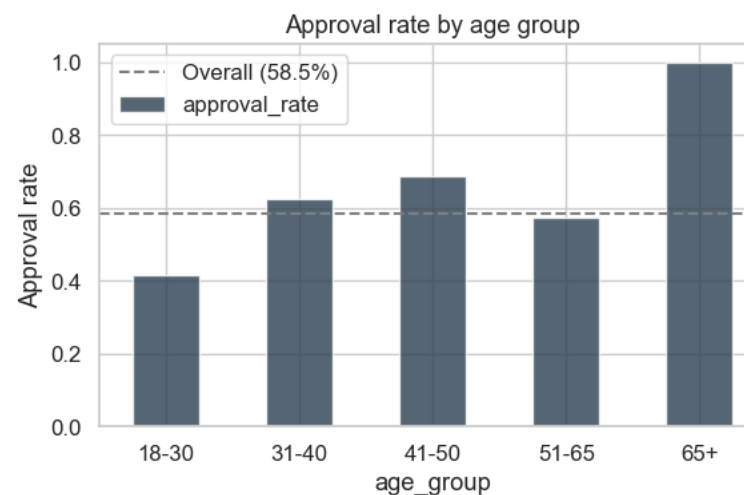
Gender Disparate Impact



- $DI = 0.767 < 0.80$: four-fifths rule violated
- Female applicants face a 15% lower approval rate
- Association is statistically significant ($p = 0.0007$)

Fairness risk: monitor DI and consider model or process review

Age-based patterns



- Segment DI (18-30 vs 41-50) ≈ 0.60 – violates four-fifths rule
- 18-30: lowest approval (~41%); 41-50 highest (~69%)
- Chi-square: approval and age group not independent ($p \approx 0.0001$).

Age-related fairness risk - prioritize 18–30 segment in review.

2 Conditional unfairness confirmed; discrimination highest for young women. Prioritize DI monitoring and segment 18 - 30

Bias & Fairness Analysis

Proxy variables

- **Geographical (ZIP/region):** Two-condition proxy test → not fully established (C1 or C2 not met); approval varies by region; female-majority region had lowest approval.
- **Financial:** Income & credit history correlate with gender and approval → proxy candidates; monitor and justify before use. Interest rates (approved only): no significant difference by gender (t-test $p \approx 0.31$) → no pricing disparity.
- **Spending behavior:** Flagged categories (Gambling, Alcohol, Adult Entertainment): two-condition test → not fully established (C1 or C2 not met).

Fairness

- **Conditional fairness:** After controlling for income, DTI, credit history, savings, age → gender still significant (OR ≈ 0.49 , $p < 0.001$); approval gap ≈ -0.16 → conditional unfairness.
- **Demographic parity ≈ 0.15** (15 pp); use for ongoing monitoring.
- **Pricing:** No evidence of interest-rate disparity by gender.

Interaction effects

- **Gender $\times$ age:** Chi-square $p \approx 0.0000$. Discrimination highest for young women: 18–30 approval $\sim 31\%$ (F) vs $\sim 53\%$ (M); segment DI ≈ 0.59 (< 0.8).
- **Gender $\times$ income:** $p \approx 0.0000$. Female approval below male in every tertile (e.g. Low $\sim 36\%$ vs 55%) → gap not explained by income.
- **Gender $\times$ ZIP/region:** Approval varies by region; region effects limited by sample size; female-majority region had lowest approval.



Recommendations

- **Monitor DI & parity** (e.g. quarterly); alert if DI < 0.8 ; re-run when model or policy changes.
- **Address conditional unfairness:** review model/process for gender or proxy use; consider fairness constraints or removing protected attributes.
- **Prioritize 18-30 & young women:** segment-specific fairness checks and mitigations.
- **Proxy monitoring:** track income/credit history; limit use of region; constrain flagged spending if used.

Eight PII fields in clear text: Pseudonymization and Age Generalization successfully Implemented. DPR and EU AI Act gaps: NovaCred credit scoring is high-risk

Privacy & Governance

PII Risk Assessment

Field	Risk	GDPR Art.
Full Name	HIGH	Art. 4(1)
Email	HIGH	Art. 4(1)
SSN	CRITICAL	Art. 87
IP Address	MEDIUM	Art. 4(1) Recital 30
Date of Birth	HIGH	Art. 4(1)
Info Gender	HIGH	Art. 4(1) / Art. 9
ZIP Code	MEDIUM	Art. 4(1)
Spending behavior	HIGH	Art. 5(1)(c)

Pseudonymization

BEFORE (plaintext)	AFTER (pseudonymised)
Name <i>Jerry Smith</i>	eb8fac8d6a30a578...
SSN <i>652-70-5530</i>	cf562a398ae20fbe...
Email <i>jerry@hotmail.com</i>	a7b3c9d1e4f2...
IP <i>192.168.1.45</i>	4d3e2f1a9b8c...
DOB <i>1987-04-12</i>	Age bracket: 31-40

Pseudonymized data retains full analytical value while meeting GDPR obligations under Art. 4(5) and Recital 26.

Further Regulatory & compliance gaps

GDPR Compliance Gaps

- Violations across 4 key principles
 - 8 GDPR provisions in gap analysis: Art. 5(1)(a), (c), (d), (e), 13/14, 17, 22, 25
- Potential fines: €20M or 4% global revenue (Art. 83)

Data subject rights

- No mechanism for access (Art. 15), rectification (Art. 16), erasure (Art. 17), portability (Art. 20), or objection (Art. 21)
- One-month response deadlines cannot be demonstrated

EU AI Act

- Credit scoring = high-risk AI system
- DI 0.77 and conditional unfairness conflict with data governance and accuracy requirements
- System not compliant until bias and high-risk obligations are addressed

3 From immediate fixes to full compliance: a phased roadmap to operationalize GDPR and EU AI Act

Privacy & Governance

PHASE 1 0–3 Months Immediate Fixes		PHASE 2 3–6 Months Structural Controls		PHASE 3 6–12 Months Full Compliance	
1	Pseudonymize SSN, name, email, IP in all systems	1	Implement consent management platform (CMP)	1	Produce EU AI Act model card and technical file
2	Enforce SSN uniqueness constraint at ingestion	2	Capture consent_timestamp + lawful_basis per application	2	Register NovaCred's system in EU high-risk AI database
3	Add processing_timestamp to every new record	3	Deploy data quality monitoring dashboard	3	Implement ongoing bias monitoring with automated alerts
4	Standardise gender encoding (M/F → Male/Female)	4	Assign Data Owner, Steward and Custodian roles formally	4	Establish data subject rights portal (access, erasure)
5	Add human_review_flag to all rejection decisions	5	Replace exact DOB with age brackets in operational store	5	Conduct independent third-party AI conformity assessment
Q1 2026		Q2 2026		Q3 2026	
				Q4 2026	

4 Four priorities: treat data quality as a legal obligation, fix bias, protect PII, and meet EU AI Act high-risk requirements.

Key Takeaways

1 Data quality is not a technical nuisance - it is a legal obligation

14 blank strings, 3 duplicate SSNs, and 87.6% missing timestamps are GDPR accuracy and accountability failures, not just engineering issues.

2 The model is algorithmically biased against women and young applicants

A Disparate Impact ratio of 0.77 (gender) and 0.60 (age 18-30) exceed the Four-Fifths legal threshold. Bias persists after controlling for income and DTI.

3 8 PII fields are at risk - and none are adequately protected today

SSN is stored in plaintext. No consent records exist. All 8 GDPR articles assessed showed compliance gaps. Pseudonymization is implementable immediately.

4 EU AI Act classification requires immediate structural action

NovaCred's credit scoring system is HIGH-RISK under Annex III. Missing model card, risk register, and human oversight mechanism expose the firm to up to 6% global revenue in fines.



Appendix

